HAND-ME-DOWN GENES

An Introduction to Genetics

Program 2: *Family Patterns*

Written by
Jan Rosenberg BEd

Edited by
David Brown and Stewart Jackel

Teacher Notes

Following many requests from teachers, these notes have been printed on A4 size sheets for easy photocopying.

111A MITCHELL STREET BENDIGO VIC 3550
TELEPHONE (03) 5442 2433 FAX (03) 5441 1148

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Program 2: Family Patterns

The two part program ***HAND-ME-DOWN GENES*** introduces basic genetic concepts for years 9 & 10 science. In particular the programs support the following curricula:
National Curriculum - Middle school science
Victorian Curriculum - CSF Life and Living strand Level 6
NSW Curriculum - Middle school science - 9 & 10

GENERAL INTRODUCTION

Almost every week there are news items of discoveries or new applications of genetic science. Genetically engineered crops and food plants, methods for detecting genetic disorders in embryos, cloned animals and plants, trials of genetic based treatments for life threatening conditions, and the production of many important drugs are just a few. It is important for students to gain a sound understanding of genetics so they can appreciate the impact of these developments and the science behind them.

These programs are designed for middle school science students and cover the fundamental ideas needed to understand how genes work and how human characteristics are inherited. They are based on the latest research in genetics and presented in such a way that students will find interesting and comprehensible. The latest in computer generated graphics and film footage of young people discussing their genetic conditions provide engaging examples of the ideas that underpin the programs.

USING THE PROGRAMS

- The programs offer explanations of the fundamental principles of genetics. The material is logical and dense and therefore lends itself to use in contained segments rather than in a single viewing.
- The programs could provide the basis for designing a whole unit of work around each of the main themes - in program 1: (1) How genes work
  in program 2: (2) Family patterns: inheritance
  (3) Genetic disorders

Program 2: FAMILY PATTERNS

Introduction:

When you get together for a family photo what do you notice? See how some of the family are very similar....but in the same moment, see how each is very different from each other! Why aren't brothers and sisters identical? And, how come some siblings ARE identical in everyway? How do we explain the sameness and diversity of family patterns?
These are the fundamental questions examined by Program 2 of *HAND-ME-DOWN GENES.*

A recap from Program 1: *How Genes Work*

We have long known that some human conditions "run in families". Like eye colour, blood group and many other characteristics, they are inherited - controlled by factors called genes. In recent years, geneticists - scientists who research inheritance- have deciphered the code system of genes and how it controls the production of all the different proteins essential for our bodies to grow and function. Program 1 describes and explains the amazing biology at work in this process with particular emphasis on DNA and the role of cell division, mitosis. Because of the duplication through mitosis, all the billions of cells in our body carry the same set of chromosomes and exactly the same alleles. The one exception is in the sex cells.

The start of inheritance: The formation of Sex Cells

The program highlights how when the body produces sex cells - what are known as the **gametes**, the cells which form sperm in males and eggs in females - the cell division process undergoes a further development from mitosis. Sex cells divide twice, first by mitosis but then, on a second division the chromosomes are not duplicated leaving resultant cells with *half* the number of original chromosomes. These go on to form sperm and eggs, or gametes, the reproductive building blocks on which each individual human being is formed.

This particular secondary cell division of the sex cells is called **meiosis.** The difference between meiosis and mitosis, and the function performed by this differential, is clearly explained and illustrated by the animated graphics and precise narration.

The Purpose of Meiosis:

So, asks the program, why should the sex cells be different from the other body cells? Well, the ultimate grand purpose is to produce new individual human beings! Here is the basis of our individual difference and the our uniqueness.
The moment of conception, when the egg is **fertilised** by the sperm, is the only time in our life cycle when two cells combine to form one cell with equal number of chromosomes from each parent. Biologically then, it is indeed a wondrous moment.

Most human body cells contain **46 chromosomes or 23 pairs**. As the gametes of each sex contain only 23 chromosomes fertilisation is necessary to restore the new cell to a total of 46 chromosomes. Once the egg is fertilised and has its full compliment of chromosomes it begins to grow and divide by mitosis.

Next question: Why differences?

Basic logic raises the next question: if chromosomes combine from the same two parents, why do children in a family show all kinds of genetic differences? And why are some male and some female?

Answer: its all in your hand-me-down genes ... different eggs and different sperm must carry different genetic information.

The program goes on to tell this part of the story starting with the way our sex is determined the sexing instructions are carried in chromosomes. Here the animated graphics clearly show how we **X and Y chromosomes** are split during meiosis and how the lottery leaves us with XX or XY. Sometimes, as we see later in the program, there are other **mutant** variations of this.

In addition we see that the chromosomes of the two parents can be reshuffled in many ways which means the games of both can carry many combinations of chromosomes . This extra dimension of 'the lottery' is the simple explanation of why the children of the same two parents show genetic differences.

There is one exception: **identical twins**, where a fertilised egg splits and two babies develop carrying exactly the same genetic information. They will be of the same sex and be truly identical. Not so for **fraternal twins** who are totally different individuals who develop from two fertilised eggs which developed at the same time in the womb.

Inheritance of Genetic conditions - Recessive and Dominant:

The remainder of the program focuses on describing and explaining a variety of "genetic conditions" where there has been a hic cup in the system and how these mishaps are inherited within families.

First we meet Madelaine, who has inherited **cystic fibrosis** where the absence of a particular protein in her system results on respiratory disorders. The protein is absent because of a **mutant allele** carried by both parents. But neither parent has the

condition because they have at least one normal allele as well as one mutant version. They are **"carriers"**. For this reason Cystic Fibrosis is called a **recessive condition**. Madelaine inherited one mutant allele from both parents and therefore the condition is expressed.

In contrast, **Huntington disease**, a disorder of the nervous system that appears late in life, is known as a dominant condition because it only requires one mutant allele for the condition to take effect.

In both these cases, a study of the family pedigree or pattern of genetic inheritance can tell us a great deal about the pathway of the condition and can help us predict when some conditions will occur. The program highlight some of the current research taking place by geneticists using the study of family patterns to extend our knowledge of inherited characteristics.

The case of Oliver, who has **achondroplasia**, underlines the fickle role of chance in the development of mutations. Achonroplasia is a dominant condition yet neither of Oliver's parents have the condition. How could this be? The explanation appears to be that a chance event during meiosis causes a slight change in the base code sequence throwing out the coding for creating a particular protein. This protein is crucial to long bone growth. So, due to a chance "spelling error" in the base coding Oliver lives with short limbs.

The program explores other forms of genetic conditions such as different eye colours and **Down Syndrome** where the mishap - an extra chromosomes on chromosome 21 - arises from an unusual event during meiosis when the egg is formed in the mother.

Mutant variation in the distribution of the sex coding X chromosome in the mother can lead to **Klinefelter Syndrome** in boys where there is an extra X chromosome (XXY). In the case **Turner Syndrome** in girls there is an X chromosome missing leaving the child with an XO pair. In both cases stature and fertility are affected. But, as an optimistic comment of the state of this science, the program subtly draws our attention to how new treatments, such as hormone therapy, are being used to successfully counteract these impacts of genetic chance.

AFTER THE PROGRAM

STUDENT ACTIVITIES

1. Create your own genetics crossword

Create clear definitions of these genetic terms in your own words and use them to create your own crossword puzzle. Test your puzzle with other students.

Chromosomes
Genes
Gametes
Fertilisation
Inheritance
Ribosome
Allele
Sperm
Egg
Geneticists
Mitosis
Meiosis
Mutation
Recessive condition
Dominant condition

2. Seeing How it Works:

a) Role play cell division in meiosis. Everyone can participate by forming nucleus and cell membranes as well as representing the dividing chromosomes.

b) Photocopy and enlarge the table on the following page. Cut the table into separate squares and jumble up the pieces. In groups ask students to put in order the processes of Mitosis and Meiosis.

3. Research.

a) Write up questions that occurred to you as you watched the program. In a small group, collate the questions on a large sheet of paper and group them into similar topic areas. Have you group present your "Questions for Further Explanation" to the class. See if the rest of the class can answer *your* questions.

b) Make a list of the genetic conditions mentioned in program one (How Genes Work) and note (1) how the condition occurs and (2) what is the effects it has on the person's development.
Research at least <u>one</u> of these genetic conditions in more detail, especially in regard to current treatments. Present your findings to the class.

4. And more creatively...

Remember the movie *Jurassic Park*? Write your own Block buster movie story but you must include these words while demonstrating their meanings.

Chromosome	Nucleus	Genes	Mitosis	Meiosis	Inheritance
DNA	Molecules	Protein	Gametes	Mutation	Fertilisation

MITOSIS	**Chromosomes in cell duplicate**	**Identical chromosome sets separate.**
Cell contents duplicate	**Cell divides to form 2 cells of equal size.**	**This process happens in body cell of plants & animals.**
MEIOSIS	**Chromosomes duplicate**	**Form two identical pairs of chromotids**
Centromere is their joining point	**4 copies of each chromosome in the cell**	**Nucleus membrane disappears**
Spindle fibres grow across the cell	**Chromosome pairs line up with spindle fibres attached to centromeres**	**Spindle fibres contract, separating members of each pair**
Each chromosome still has 2 chromatids.	**Division occurs again. Each cell now has half chromosomes of original cell**	**These cells are either sperm or egg.**
Occurs in body cells	**Occurs in sex cells**	

OTHER RELATED PROGRAMS AVAILABLE ON 14 DAY PREVIEW FROM VEA

- ***The story of the Gene***
- ***Modern Genetics***
- ***Gene Genius***
- ***Meiosis - The key to genetic Diversity***

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